

Excel + Power BI — Task 2

Advanced KPI Analysis & Interactive Business Dashboard

Prepared by	Business Analytics Intern
Organisation	Maincrafts Technology
Dataset	Sales Data — 200 Orders (FY 2024)
Tools Used	Microsoft Excel + Power BI Desktop
Submitted	June 2026

01. KPI Definitions

Key Performance Indicators (KPIs) translate raw data into actionable business metrics. The following KPIs were defined and calculated for this dashboard:

\$842,350 Total Sales Sum of all order revenues across regions, products, and time periods. Primary revenue indicator.	\$198,720 Total Profit Net earnings after deducting costs from sales. Measures business profitability.	23.6% Profit Margin Profit as a percentage of Sales. Indicates how efficiently revenue converts to profit.
200 Total Orders Count of unique transactions. Measures sales volume and business activity.	2,145 units Total Quantity Total units sold. Highlights demand patterns across products.	\$4,212 Avg Order Value Average revenue per transaction. Higher values indicate premium customer segments.

KPI Formula Reference

KPI	Excel Formula	DAX Measure (Power BI)	Purpose
Profit Margin	=Profit/Sales	DIVIDE([Total Profit],[Total Sales])	Profitability ratio
Sales Category	=IF(Sales>5000,"High","Medium")	Calculated column via IF	Segment high-value orders
Total Sales	=SUM(Sales column)	SUM(SalesData[Sales])	Total revenue
Total Profit	=SUM(Profit column)	SUM(SalesData[Profit])	Net earnings
Monthly Sales	Pivot Table (Month)	CALCULATE([Total Sales], VALUES(OrderDate))	Time-based trend
YTD Sales	N/A	TOTALYTD([Total Sales], SalesData[Order Date])	Year-to-date cumulative

02. Key Insights

The following insights were derived from analysing 200 sales orders across 4 regions, 3 product categories, and 12 months of FY 2024 data. Each insight is grounded in the KPI metrics and dashboard visualisations.

01	Revenue vs. Profit Imbalance High sales volumes in Electronics do not always yield high profit margins. Several high-ticket electronic items showed margins below 12%, dragging the overall category margin down despite contributing the largest share of total revenue. Recommendation: Review supplier costs and adjust pricing strategy for low-margin electronic SKUs. A 3-5% margin improvement across this category could add ~\$25K in annual profit.
02	Regional Performance Disparity One region consistently generates the highest profit-per-order despite ranking third in total order volume. This indicates a high-value customer base with premium purchasing behaviour — a pattern often missed when looking at volume metrics alone. Recommendation: Focus account management resources on this region. Launch a loyalty or key-account programme to retain and expand these high-value customers.
03	Seasonal Sales Peaks in Q4 Monthly trend analysis reveals a clear spike in sales during October through December, with November recording the highest single-month revenue. This seasonal pattern is consistent across all regions and product categories. Recommendation: Pre-position inventory and launch targeted marketing campaigns from September. Budget 30-40% of annual promotional spend for Q4 to maximise peak-season returns.
04	Office Supplies — Volume Leader, Margin Laggard Office Supplies drives the highest number of individual orders (volume leader) but contributes the lowest average revenue per order. Profit margins in this category are compressed due to high competition and low unit prices. Recommendation: Introduce bundle offers (e.g., Pen Set + Notebook + Stapler) to increase average order value. Cross-sell premium variants alongside standard products.

05

High-Sales, Low-Margin Products Need Attention

Products classified as High Sales (Sales > \$5,000 per order) but showing profit margins below 15% represent a critical risk. These products consume fulfilment resources disproportionate to the profit they generate.

Recommendation: Audit the top 10 high-sales products by margin. Negotiate better supplier rates or implement a minimum margin threshold policy for bulk orders.

06

Quantity vs. Revenue Mismatch in Furniture

Furniture items sell in moderate quantities but generate disproportionately high revenue per transaction, making them a high-value, low-volume category. However, return rates and delivery costs in this category can erode margin significantly.

Recommendation: Upsell extended warranties or complementary products at point of sale. Negotiate logistics contracts to reduce per-unit delivery costs for bulky furniture items.

03. Business Recommendations

Based on the KPI analysis and insights above, the following strategic recommendations are presented for leadership action. Each recommendation includes a priority rating and expected business impact.

HIGH	1. Margin Recovery Programme Conduct a full margin audit of all products with Profit Margin below 15%. Renegotiate supplier contracts for the bottom 20% of margin performers. Set a company-wide minimum acceptable margin threshold of 15% for new product listings. Expected impact: 2-4% improvement in overall profit margin within 2 quarters.
HIGH	2. Q4 Revenue Maximisation Develop a dedicated Q4 sales playbook leveraging the observed seasonal peak. Pre-load inventory by mid-September, launch promotional campaigns in early October, and set aggressive targets for the November peak window. Expected impact: 15-20% uplift in Q4 revenue compared to Q3.
MEDIUM	3. Regional Key Account Strategy Identify the top 10 customers in the highest profit-per-order region and assign dedicated account managers. Create a tiered loyalty programme offering volume discounts and priority service. Expected impact: 10% increase in repeat order rate from this region.
MEDIUM	4. Product Bundling for Office Supplies Design 3-5 curated product bundles for Office Supplies to raise the average order value from this high-volume, low-revenue category. Test bundle promotions with a 10% discount and measure uplift in average transaction size. Expected impact: 20-30% increase in Office Supplies revenue per order.
LOW	5. Dashboard-Driven Monthly Reviews Institutionalise monthly management reviews using the Power BI dashboard. Track the 6 core KPIs against monthly targets, highlight regions or products deviating more than 10% from plan, and trigger corrective actions. Expected impact: Faster response to performance gaps, improved forecast accuracy.

04. Dashboard Visuals — Power BI Reference

Visual	Type	X-Axis / Legend	Y-Axis / Value	Purpose
KPI Cards	Card	—	Total Sales, Profit, Margin	At-a-glance snapshot
Monthly Sales Trend	Line Chart	Order Date (Month)	Monthly Sales measure	Spot seasonality

Profit by Region	Bar Chart	Region	Total Profit	Compare regions
Top 10 Products	Table	Product	Sales, Profit, Margin	Rank performers
Category Share	Pie Chart	Category	Total Sales	Category contribution
Sales Distribution	Scatter	Quantity	Profit Margin	Identify outliers

This report was produced as part of Maincrafts Technology Business Analytics Internship — Task 2. All data is synthetic and generated for learning purposes only.